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・等差型 $a_{n+1} = a_n + d$, 等比型 $a_{n+1} = r a_n$, 階差型 $a_{n+1} = a_n + f(n)$

① $a_{n+1} = a_n + \underline{2}$, $a_1 = \underline{1}$ $\Rightarrow 1 a_{n+1} = 1 a_n + \underline{d}$ (等差型)

$$a_n = 1 + (n-1) \cdot 2$$

$$= 2n - 1$$

$$\left. \begin{array}{l} a_1 = 1 \\ a_2 = a_1 + 2 = 3 \\ a_3 = a_2 + 2 = 5 \\ a_4 = a_3 + 2 = 7 \\ \vdots \end{array} \right\} \begin{array}{l} \textcircled{1} \textcircled{2} \textcircled{3} \textcircled{4} \\ \text{の等差数列} \end{array}$$

② A $a_{n+1} = \underline{2} a_n$, $a_1 = \underline{1}$ $\Rightarrow 1 a_{n+1} = \underline{r} a_n$ (等比型)

$$a_n = 1 \cdot 2^{n-1}$$

$$= 2^{n-1}$$

$$\left. \begin{array}{l} a_1 = 1 \\ a_2 = 2a_1 = 2 \\ a_3 = 2a_2 = 4 \\ a_4 = 2a_3 = 8 \\ \vdots \end{array} \right\} \begin{array}{l} \textcircled{1} \textcircled{2} \textcircled{3} \textcircled{4} \\ \text{の等比数列} \end{array}$$

③ $a_{n+1} = a_n + \underline{2^n}$, $a_1 = 1$ $\Rightarrow 1 a_{n+1} = 1 a_n + \underline{f(n)}$ (階差型)

$n \geq 2$ のとき

$$a_n = a_1 + \sum_{k=1}^{n-1} 2^k$$

$$= 1 + 2^1 + 2^2 + \dots + 2^{n-1}$$

$$= 2^n - 1$$

ここで、初項は $a_1 = 1$ なので

① は $n=1$ のときも成り立つ。

よって

$$a_n = 2^n - 1$$

$$a_{n+1} - a_n = 2^n$$

$$a_2 - a_1 = 2^1$$

$$a_3 - a_2 = 2^2$$

$$a_4 - a_3 = 2^3$$

$$\vdots$$

$$+) a_n - a_{n-1} = 2^{n-1}$$

$$a_n - a_1 = 2^1 + 2^2 + \dots + 2^{n-1}$$

$$a_n = a_1 + \sum_{k=1}^{n-1} 2^k \quad (n \geq 2)$$

point

① 等差型 $1 a_{n+1} = 1 a_n + \underline{d}$ $\Rightarrow a_n = a_1 + (n-1) \cdot d$

② 等比型 $1 a_{n+1} = \underline{r} a_n$ $\Rightarrow a_n = a_1 \cdot r^{n-1}$

③ 階差型 $1 a_{n+1} = 1 a_n + \underline{f(n)}$ $\Rightarrow a_n = a_1 + \sum_{k=1}^{n-1} f(k) \quad (n \geq 2)$